

Xiaoyu Lyu

M: (86) 18563806108 | E: xiaoyu.lyu01@gmail.com

Address: 222 Tianshui South Road, Chengguan District, Lanzhou, Gansu Province 730000, P.R. China

EDUCATION

Lanzhou University, College of Ecology
Master of Science in Ecology
Weighted Average: 87.3/100

Lanzhou, China
2024.9 – Present

Northeast Forestry University, College of Landscape Architecture
Bachelor of Engineering in Landscape Architecture
Weighted Average: 85.17/100

Harbin, China
2020.9 – 2024.6

RESEARCH INTERESTS

Cereal–Legume Intercropping Rhizosphere Microbiome Nutrient Turnover Spatial Agroecology (GIS/MaxEnt)

RESEARCH EXPERIENCE

Graduate Research Assistant | Lanzhou University, College of Ecology

2025.4 – Present

Project: Driving Effects of Maize-Legume Intercropping on Nutrient Turnover in Dryland Agroecosystems

- **Dryland Field Trial Design & Management:** Set up and managed large-scale rainfed field experiments to evaluate interspecific competition and resource sharing. Designed multiple intercropping treatments pairing maize with two distinct legumes: lablab beans and forage soybeans. This comparative setup aimed to test how different legume species interact with the main crop and compete for limited precipitation and space.
- **Soil Moisture & Nutrient Turnover Monitoring:** Tracked root-zone soil moisture dynamics across different sowing periods under drought stress. Sampled and quantified belowground carbon, nitrogen, and phosphorus turnover rates to assess soil fertility changes. Specifically analyzed the severe resource competition within these systems, focusing on the nutrient and water dynamics that led to significantly reduced maize grain yields, particularly in the lablab intercropping treatments.

Graduate Research Assistant | Lanzhou University, College of Ecology

2026.6 – Present

Project: Isolation and Cultivation of Halo-Alkalitolerant Microorganisms from Saline Soils

- **Field Sampling & Stress-Gradient Screening:** Conducted fieldwork to collect soil samples from naturally occurring saline-alkali environments across Northwest China. Prepared soil suspensions and plated them onto specialized selective media. By applying varied gradients of salinity and pH, I systematically screened for robust bacterial strains capable of surviving under dual halo-alkaline stress conditions.
- **Pure Culture & Microbiological Workflows:** Achieved pure cultures through rigorous and repeated streak-plate isolation and subculturing protocols. Routinely executed standard laboratory workflows, emphasizing strict aseptic techniques, phenotypic colony characterization, and the long-term cryopreservation of these extremophile strains for downstream functional analysis.

Graduate Research Assistant | Lanzhou University, College of Ecology

2025.10 – 2025.12

Project: Habitat Suitability Modeling and Spatial Zoning Using MaxEnt and GIS

- **Ecological Niche Modeling:** Constructed predictive models via MaxEnt to evaluate the environmental suitability for specific target crops. Combined regional bioclimatic datasets with detailed soil physicochemical properties to identify the primary environmental drivers of crop distribution and to establish critical thresholds for plant survival.
- **Spatial Analysis & Cultivation Zoning:** Processed the modeling results within a GIS environment to perform spatial overlay analyses and generate high-resolution suitability maps. Applied these spatial predictions to classify land into distinct agricultural zones, providing a concrete spatial framework for crop placement and local land-use planning.

SKILLS AND INTERESTS

- **Computational & Analytical Skills:** Proficient in R for statistical evaluation and data visualization, alongside strong spatial analysis capabilities using QGIS/ArcGIS. Leveraging my background in Landscape Architecture, I utilize AutoCAD for detailed site planning diagrams, and Adobe Photoshop for conceptual schematics and publication-ready figure assembly.
- **Laboratory & Instrumental Techniques:** Fully competent in standard microbiology lab operations, including strict aseptic workflows, microbial inoculation, and routine sterilization procedures. Expanding analytical skill set through upcoming hands-on training with Continuous Flow Analyzers (CFA) and spectrophotometers for precise soil nutrient and enzymatic assays.
- **Personal Interests:** Strongly interested in open-source software environments (e.g., Linux ecosystems) and the design of sustainable agricultural systems. Outside the lab, I enjoy nature hiking and botanical field observation, which keeps me grounded in my broader passion for soil-plant interactions and agrobiodiversity.